

LOW COST SENSOR BEHAVIOUR ASSESMENT: DEALING WITH LONG-TERM DRIFTS

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Summary

A low-cost sensor, Kunak-Air 10, assessment is performed in the study. Its behaviour is evaluated and compared, in four different situations, with a fixed and a mobile reference station, as well as with another Kunak-Air 10, during a period of one year. It is concluded how this type of sensors have good performance, with an accuracy a precision near to references stations.

Introduction

Regarding the World Meteorological Organization (WMO), the main challenges to deal with using low-cost sensors (LCSs) are (1) the high temperature and relative humidity values, (2) the cross-sensitivity within some atmospheric compounds and (3) the long-term stability and drifts. The objective of the paper is to evaluate the behaviour of a specific LCS, Kunak-Air 10 (K-Air 10) (<https://www.kunak.es/>) under different field conditions, in Mexico City from Jan 2019 to Dec 2020. This study is part of a project which covers other studies in five continents. The World Athletics and UN Environment (UNEP) are addressing together the poor air quality issue, which has led to seven million deaths globally, according to a World Health Organization study. Before Mexico City, Kunak Technologies S.L has been integrating air quality monitors in several stadiums around the world, in Monaco, Addis Ababa, Sydney and Yokohama.

Methodology and Results

Kunak-Air 10 version 2 provided with CO, NO₂, NO, O₃, NO_x, PM₁₀, PM_{2.5}, PM₁, as well as temperature, pressure, humidity and wind sensors. Two K-Air 10 V2 were set in Mexico City, K-Air 10 V2 #1 and K-Air 10 V2 #2. Both were factory calibrated prior to the study and then, they were placed in two different locations: K-Air 10 V2 #1 in the Mexican Olympic Committee (COMex) and K-Air 10 V2 #2, on top the Cien Metros Reference Station. In this context, the study was set in four difference phases, in which K-Air 10 V2 #2 behaviour was assessed. Phase I: K-Air 10 V2 #2 sensor was set and compared with the Cien Metros Fixed reference station and its baseline and span was slightly adjusted. Phase II: K-Air 10 V2 #2 was moved to another location co-located to mobile UMO reference station during CDMX Marathon without any further re-calibration, in order to be compared with the reference station. Phase III, K-Air 10 V2 #2 location was changed to COMex with K-Air 10 V2 #1. In this phase, K-Air 10 V2 #2 was used to adjust K-Air 10 V2 #1 drifts, which has not moved from its original position since January 2019. The data of K-Air 10 V2 #2 was averaged every hour to match data from the reference stations and the 100% of the data was used for the analysis. For instance, it is obtained correlation values of $R^2 = 0.95$ and 0.97 for CO concentration for K-Air 10 V2 #2 with the Cien Metros fixed reference station, and with the mobile UMO reference station.

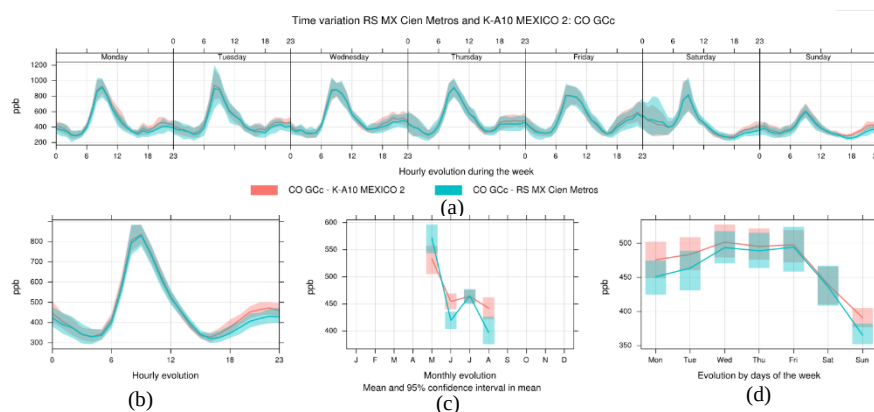


Figure 1. Phase I of K-Air 10 V2 #2 and the Cien Metros Reference Station: CO hourly evolution during (a) the week and (b) the day. Mean and 95% confidence interval in mean for (c) monthly and (d) weekly evolution.

Conclusions

After analysing the data, it is possible to conclude how LCSs can have a very good performances being very useful to extend air quality reference networks. Specifically, Kunak-Air 10 is able to measure with the accuracy and precision very close to reference station. Moreover, the long-term stability and drifts issue is solved by controlling the sensor remotely. Additionally, a procedure to guarantee the quality check and quality assurance is studied by moving the sensor and verifying its behaviour, checking how the precision and accuracy are maintained in time. Finally, Phase III allows to estimate the drift, and thus, it is possible to give some recommendations about the calibration durability, as well as using the reference stations as a QC&QA element for a specific sensor and for the measurement's campaigns.

References

Lewis, A., Peltier, W. R., & von Schneidmesser, E. 2018. Low-cost sensors for the measurement of atmospheric composition: overview of topic and future applications.